

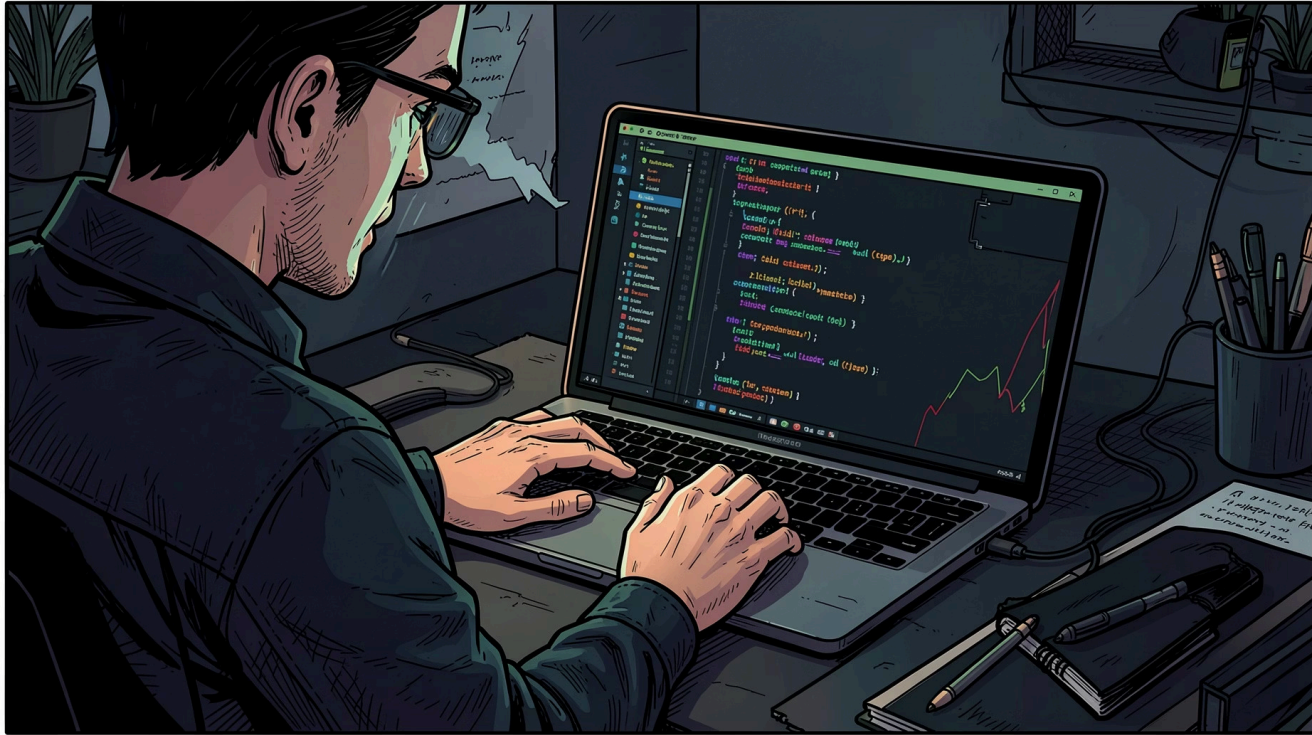


ANALYSIS OF GLOBAL COVID-19 PANDEMIC DATA

Using Wikipedia and R to collect, clean, and interpret global testing and confirmed case data – revealing patterns in infection prevalence, testing coverage, and country-level comparisons.

R · WEB SCRAPING · PUBLIC HEALTH ANALYTICS

INTRODUCTION & OBJECTIVES



The COVID-19 outbreak became one of the most significant global health crises in modern history. Governments relied on testing data to monitor spread and implement containment measures.

THIS PROJECT DEMONSTRATES HOW TO:

- Retrieve COVID-19 testing data from Wikipedia via HTTP requests
- Clean and preprocess the dataset
- Calculate global testing statistics and positivity rates
- Compare infection prevalence between countries

TOOLS & TECHNOLOGIES



R & RSTUDIO

Core language and development environment for all analysis



HTTR & RVEST

HTTP requests and web scraping to extract HTML table data



DPLYR & XML2

Data manipulation and HTML parsing



GGPLOT2

Data visualization for charts and graphs



WIKIPEDIA

Primary data source – COVID-19 testing page

DATA COLLECTION & PREPROCESSING

HTTP REQUEST

An HTTP GET request to the Wikipedia COVID-19 testing page returned status code **200**, confirming successful server communication. HTML content was parsed and converted into a structured data frame using `rvest`.

```
library(httr)
url <- "https://en.wikipedia.org/wiki/COVID-19_testing"
response <- GET(url)
status_code(response)
```

DATA CLEANING STEPS

Removed duplicate rows
and missing values

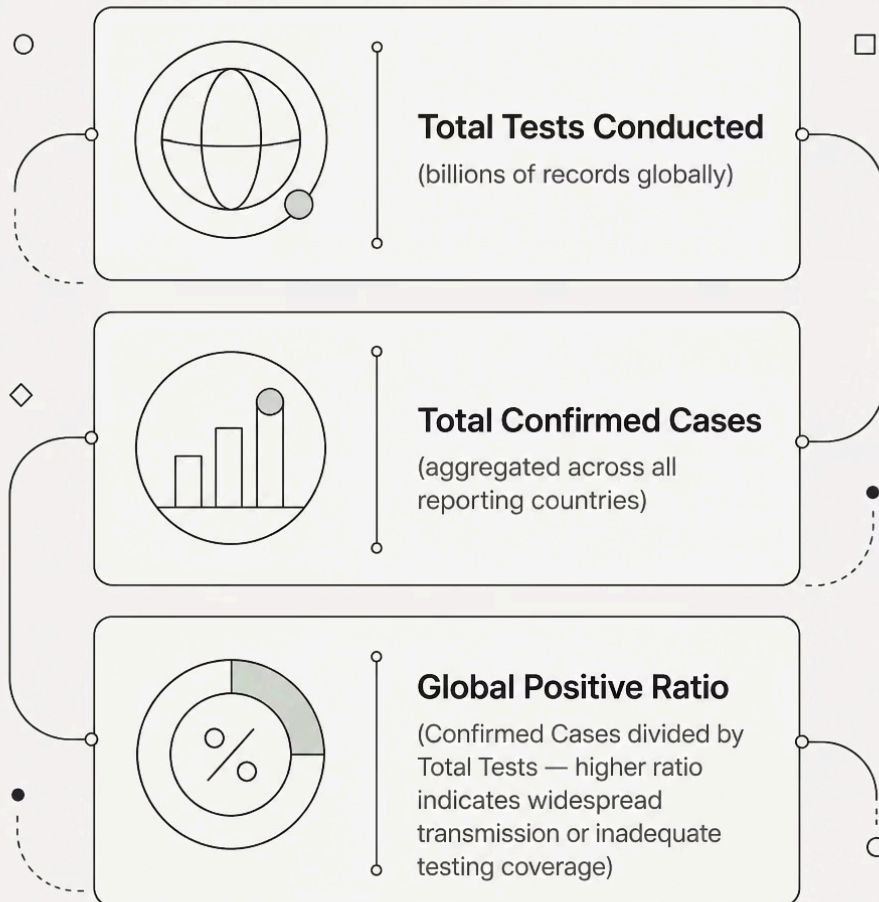
Eliminated Wikipedia
reference tags

Removed commas from
numerical values

Converted text fields to
numeric format

Standardized country names

EXPLORATORY DATA ANALYSIS



POSITIVE RATIO FORMULA

$$\text{Positive Ratio} = \text{Total Confirmed Cases} \div \text{Total Tests Conducted}$$

A higher positivity ratio may indicate widespread transmission or inadequate testing coverage. This metric served as a key indicator of global infection prevalence.

-  Global testing efforts generated billions of test records, highlighting the massive scale of pandemic monitoring worldwide.

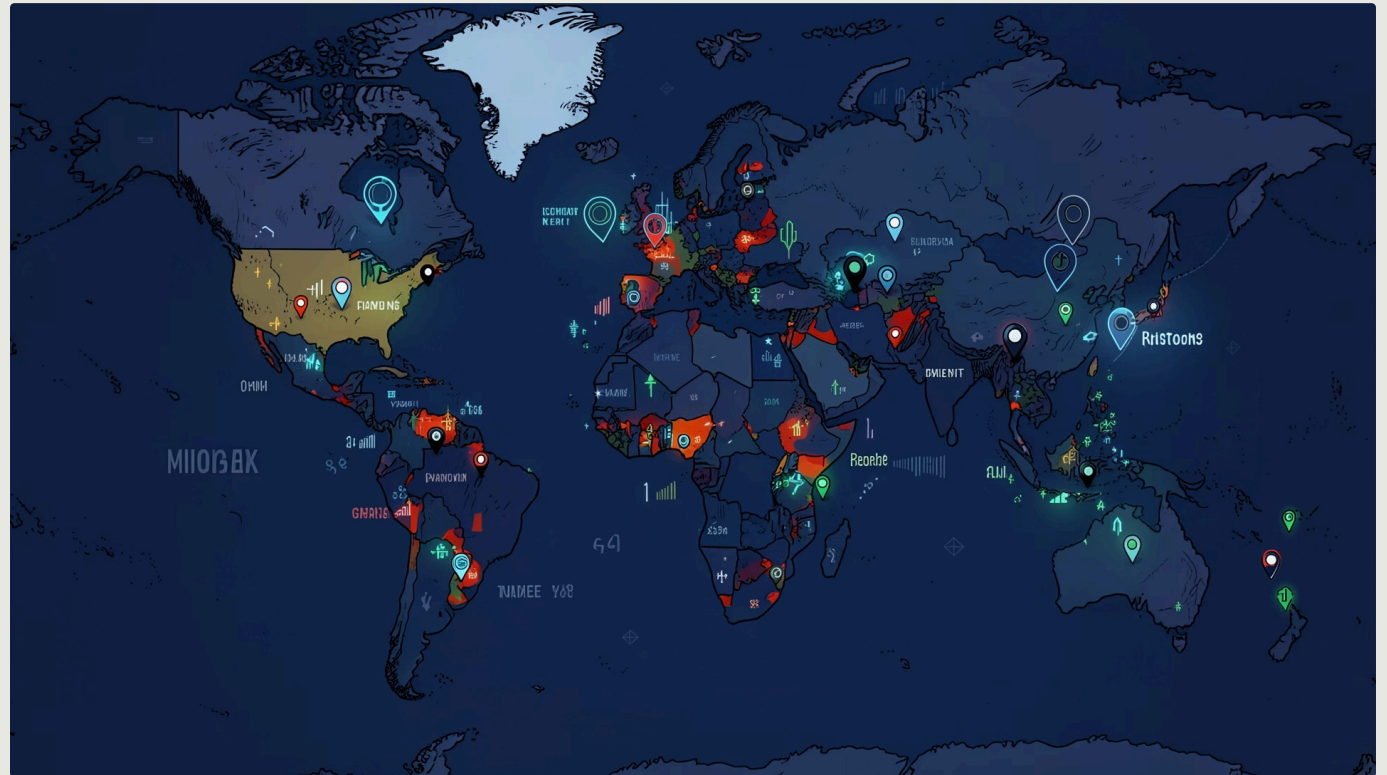
COUNTRY REPORTING & PATTERN RECOGNITION

REPORTING FINDINGS

Hundreds of countries and territories reported testing information, though reporting frequency varied significantly across regions. Data availability reflected differences in public health infrastructure and reporting practices.

REGULAR EXPRESSION MATCHING

Regex was used to identify countries beginning with **"United"**, demonstrating pattern matching in data analysis:



UNITED STATES

UNITED KINGDOM

UNITED ARAB EMIRATES

COUNTRY COMPARISON: USA VS. INDIA



METRICS ANALYZED

The comparison revealed **substantial differences** in infection prevalence relative to population size. Countries with higher confirmed-to-population ratios experienced a greater reported burden of COVID-19 infections.

CONFIRMED CASES

Total reported infections per country

CONFIRMED-TO-POPULATION RATIO

Confirmed Cases ÷ Population – a normalized measure of infection burden

INFECTION RISK ASSESSMENT

CONFIRMED-TO-POPULATION RATIO

Formula: Confirmed Cases ÷ Population

Countries with larger ratios exhibited a greater proportion of their population reported as infected. This metric served as the primary indicator for comparing reported infection risk across countries of different sizes.

- This normalized measure allows fair comparison between countries regardless of population size.

LOW-RISK COUNTRY IDENTIFICATION

Countries with a confirmed-to-population ratio **below 1%** were flagged as having relatively low reported infection prevalence. Potential reasons include:

Effective containment measures & early intervention

Geographic isolation & lower population density

Differences in testing and reporting policies

KEY INSIGHTS

1

SCALE OF TESTING

Billions of test records were generated globally, highlighting the massive scale of pandemic monitoring.

2

POSITIVITY RATIO

The worldwide positivity ratio provides a useful indicator of global infection prevalence.

3

COUNTRY VARIATION

Substantial variation exists between countries in testing rates and confirmed case ratios.

4

NORMALIZED COMPARISON

Confirmed-to-population ratios offer a fair measure for comparing infection prevalence across different population sizes.

5

LOW-RISK THRESHOLD

Countries below 1% may have lower reported infection rates, though testing intensity must also be considered.

CHALLENGES, CONCLUSION & FUTURE WORK

CHALLENGES

Dynamic Wikipedia content changes over time

Missing values and inconsistent formatting

Different reporting methodologies across countries

Potential under-reporting or delayed reporting

CONCLUSION

This project demonstrated how web scraping and data analysis in R can extract real-world pandemic data and derive meaningful insights on global testing efforts and infection prevalence.

FUTURE ENHANCEMENTS

- Build interactive dashboards using Shiny
- Automate daily data collection
- Perform time-series forecasting
- Integrate multiple public health datasets
- Develop ML models for outbreak prediction